⌈Thm. If $f$ is a continuous complex-valued function on $[a,b]$, then there exists a sequence of polynomials $P_n$ such that

$$P_n \to f \text{ uniformly on } [a,b].$$

⌈Proof. WLOG, we assume $[a,b]=[0,1]$, and $f(0)=f(1)=0$.

(This is possible: $g(x)=f(x)-f(0)-x(f(1)-f(0))$)

And, $f(x)=0$ if $x \notin [0,1]$.

Put $Q_n(x)=C_n\cdot(1-x^2)^n \quad (n=1,2,3,\cdots)$

where $C_n$ is chosen such that $\int_{-1}^{1} Q_n(x)\,dx=1$.

To magnitude $C_n$,

$$\int_{-1}^{1}(1-x^2)^n\,dx = 2\cdot\int_0^1 (1-x^2)^n\,dx \geq 2\cdot\int_0^{\frac{1}{\sqrt{n}}}(1-x^2)^n\,dx$$

$(1-x^2)^n-(1-nx^2)$

$\downarrow\ n(1-x^2)^{n-1}\cdot(-2x)+2nx$

$$\geq 2\cdot\int_0^{\frac{1}{\sqrt{n}}}(1-nx^2)\,dx \qquad =2nx\cdot(-(1-x^2)^{n-1}+1)\geq 0.$$

$$=2\cdot\left(\frac{1}{\sqrt{n}}-\frac{n}{3}\cdot\frac{1}{(\sqrt{n})^3}\right)=\frac{4}{3}\frac{1}{\sqrt{n}}>\frac{1}{\sqrt{n}}$$

So, $C_n<\sqrt{n}$. Now, for any $\delta>0$,

$Q_n(x)\leq\sqrt{n}\cdot(1-\delta^2)^n \quad (\delta\leq|x|\leq 1)$, so that $Q_n\to 0$ uniformly in $\delta\leq|x|\leq 1$

Now, set $P_n(x)=\int_{-1}^{1} f(x+t)\cdot Q_n(t)\,dt \quad (0\leq x\leq 1)$.

Since $f(s)=0$ if $s\notin[0,1]$, (Revised in the next page)

$$=\int_{-x}^{1-x} f(x+t)\,Q_n(t)\,dt$$

$$=\int_0^1 f(s)\,Q_n(s-x)\,ds \quad (x+t=s \Rightarrow dt=ds)$$

Given $\varepsilon>0$, choose $\delta>0$ s.t. $|x-y|<\delta \Rightarrow |f(x)-f(y)|<\frac{\varepsilon}{2}$, since $f$ is continuous on compact

$$|P_n(x)-f(x)|=\left|\int_0^1[f(x+t)-f(x)]\cdot Q_n(t)\,dt\right|\leq\int_0^1|f(x+t)-f(x)|\cdot Q_n(t)\,dt$$

$$\leq 2M\cdot\left(\int_{-1}^{-\delta}Q_n(t)\,dt+\int_{\delta}^{1}Q_n(t)\,dt\right)+\frac{\varepsilon}{2}\cdot\int_{-\delta}^{\delta}Q_n(t)\,dt \quad (M=\sup|f|)$$

$$\leq 4M\cdot\sqrt{n}\,(1-\delta^2)^n+\frac{\varepsilon}{2}\to 0$$